

Drexel University, Philadelphia PA

Bachelor of Science in Computer Science; Minor in Data Science and Financial Technology

TECHNICAL SKILLS

Graduating June 2027

GPA: 3.8 - Dean's List

Languages: Python, Java, SQL, JavaScript, HTML, CSS, C, C++, Rust, Bash

Libraries: NumPy, Pandas, Matplotlib, SQLAlchemy, jQuery, Bootstrap, Thymeleaf, Streamlit, FastAPI, ReactJS, Tokio

Frameworks/ Runtimes: Spring Boot, LangChain, LangGraph, Node.js, TailwindCSS, JUnit, OAuth 2.0

Databases: Microsoft SQL Server, PostgreSQL, PGVector, SQLite, Redis

Software Development: Agile Methodologies, Git, Gitflow, Large Language Models, System Design, RESTful API, Docker

RESEARCH EXPERIENCE

Drexel University, College of Computing and Informatics – Philadelphia, PA

Student Research Assistant - Software Engineering and Analytics Research (SOAR) Lab

- Owned end-to-end development of **TraceView**, an interactive visualization tool for agentic program repair trajectories; developed **Python**, **Jupyter Notebook**, and **Streamlit** tooling for trajectory annotation, graph-based visualization, filtering, metrics, and failure analysis, and contributed to the paper's methodology section by documenting the system design and implementation.

- Contributed to **AXE**, a multi-agent exploit engine for confirming zero-day vulnerability reports, supporting agentic vulnerability validation and exploitability analysis; co-authored the **AXE** paper under review and am contributing to a fourth manuscript in preparation

Drexel University, College of Computing and Informatics – Philadelphia, PA

Student Research Assistant

- Developed a simulation framework for InterPlanetary Archival Record Objects (**IPAROs**) in **IPFS/IPNS**, using **Bash** automation and Python (**NumPy**, **Pandas**) to evaluate 11 decentralized linking strategies through large-scale benchmarking

- Built data collection and visualization pipelines using **Matplotlib** to analyze storage efficiency, retrieval time, and scalability of IPARO implementations across thousands of simulated versions, validating theoretical complexity bounds with empirical results; presented findings at **JCDL 2025**

SELECTED PUBLICATIONS

- Patrick Le, Quang Bui, Alexander Grigorian, Alexey Kuraev, Thiyazan Salman, **Tu H. Nguyen**, Mat Kelly, Sawood Alam. "[Linking Policies for the Permanent Web](#)." JCDL 2025. DOI: 10.1109/JCDL67857.2025.00033

- Amirali Sajadi, **Tu H. Nguyen**, Kimmie Huynh, Esteban Parra, Preetha Chatterjee. "[TraceView: Interactive Visualization of Agentic Program Repair Trajectories](#)." ICSME 2026 (accepted).

- "[AXE: An Agentic eXploit Engine for Confirming Zero-Day Vulnerability Reports](#)." Under review.

PROJECTS

Super Streaming: Low-latency Pub/Sub System

June–August 2024

- Designed and developed a high-performance pub/sub system in **Rust** using **Tokio**, achieving **sub-millisecond end-to-end latency** with up to 1k clients and sub-30ms latency with up to 10k clients, while ensuring data throughput of up to **500,000 messages/second**

- Created a multithreaded, MQTT v5-inspired system, purpose-built for streaming stock data from two exchanges, optimized for performance and scalability

iAgent : AI-Driven Project Assistance Tool

April–June 2024

- Designed and developed a full-stack internal application using **FastAPI**, **SQLAlchemy**, **PostgreSQL**, **React**, and **TailwindCSS** to support AI-assisted project planning and document workflows

- Built core platform features including a responsive multi-page frontend, REST API, **PGVector**-based vector store, and user authentication via **JWT** and **OAuth 2.0**

- Implemented a document management and **RAG** workflow using **OpenAI embeddings**, enabling users to organize workspace files and generate project requirements, test cases, and market research outputs

WORK EXPERIENCE

Drexel University Information Technology, Core Enterprise Systems – Philadelphia, PA

Junior Developer

- Engineered new features and microservices for the **DrexelOne** portal, a production system serving thousands of students, faculty, and staff; implemented scalable integrations with **Microsoft SQL Server** using **Java**, **Spring Boot**, and **REST APIs**

- Built responsive web interfaces and faculty-facing applications using **jQuery**, **Bootstrap**, and **Thymeleaf**, translating specifications into production features used across university systems

SSI Securities Corporation – Ho Chi Minh City, Vietnam

Software Engineer Intern

- Designed and developed various internal tools leveraging **Generative AI** to enhance customer care, streamline operations, and accelerate project development

- Engaged with cross-functional teams to diagnose and rectify critical issues in active projects using the **Gitflow** branching model, leading to streamlined workflows and enhanced project efficiency

TECHNICAL INVOLVEMENT